Pharmaceutical Synthesis and Characterisation

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical Synthesis and Characterisation (A07736)

Short Title:	Pharm Synth & Characterisation
Faculty	Science and Computing
Department	Science
Cluster	Chemistry
Author	CLENNON
Credits	5

Level: Advanced

Description of Module / Aims

This module will cover the role of spectroscopic methods, specifically nuclear magnetic resonance (NMR), for the structural elucidation of organic compounds. A review of functional group transformations and carbon-carbon bond forming reactions will be undertaken, with a focus on evaluating routes to synthesise pharmaceuticals. Additional techniques for the design of efficient selective organic syntheses will be presented.

Programmes

CHEM-0032	BSc (Hons) in Applied Chemistry with Quality Management (WD_SQMCH_B)	4 / 1 / M
PHAR-0017	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	4 / 7 / M

Indicative Content

- Advanced Spectroscopy: fourier transform NMR techniques such as decoupling and the nuclear Overhauser effect (NOE); multi-pulse techniques such as DEPT; two-dimensional NMR spectra; use of expert systems
- Application and detailed analysis of advanced spectroscopic techniques for the identification of complex organic and pharmaceutical compounds
- A comprehensive review of functional group transformations, carbon-carbon and carbon-nitrogen bond forming reactions; application of such reactions to the synthesis of targeted pharmaceuticals along with relevant precursors such as heterocyclic components.
- Organometallic reactions, reagent formation and selectivity in reactions, Palladium catalysed carbon-carbon bond formation including Heck, Suzuki and Stille reactions, regio- and stereochemical aspects are discussed.
- Design aspects of synthetic routes, retrosynthesis; importance of chemoselectivity, regioselectivity and protecting groups
- Research-based examples of advanced organic synthesis and application of synthesised compounds, e.g. supramolecular compounds, organocatalysts

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Propose and design synthetic routes to targets by applying an in-depth knowledge of organic reactions.
2. Interpret such routes towards the syntheses of pharmaceuticals by applying knowledge of functional group transformations and carbon-carbon and carbon-nitrogen bond forming reactions.
3. Compare the various C-C forming reactions which utilise organometallic reagents and describe in detail reagent formation and selectivity. Distinguish between Suzuki, Heck and Stille palladium catalysed reactions.
4. Distinguish issues of chemoselectivity and regioselectivity when designing or interpreting synthetic routes.
5. Determine complex product structure by applying a thorough knowledge of advanced nuclear magnetic resonance (NMR) spectroscopy and its applications.

Learning and Teaching Methods

- Lectures.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5

Assessment Criteria

<40%: Inadequate understanding of pharmaceutical synthesis and characterisation at even a basic level.

40%-49%: Will be able to show a basic knowledge of pharmaceutical synthesis and characterisation.

50%-59%: As well as the above, the student will be able to demonstrate an understanding of some of the complexities of pharmaceutical synthesis and characterisation and predict trends in characterisation results.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of pharmaceutical synthesis and characterisation and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material.

70%-100%: All of the above to an excellent level. The learner will be able to demonstrate comprehensive knowledge and understanding of pharmaceutical synthesis and characterisation, predict general trends in behaviour, including exceptional behaviour and critically evaluate and relate topics.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- Clayden, J., N. Greeves and S. Warren. _Organic Chemistry_ . 2nd ed. : Oxford University Press, 2012.

- Sanders, J.K.M. _Modern NMR Spectroscopy_. Oxford: Oxford University Press, 1993.

Supplementary Material

- Lednicer, D. _Organic Chemistry of Drug Synthesis_. Hoboken: Wiley, 2008.
- Meakins, D.G. _Functional groups : characteristics and interconversions_. Oxford: Oxford University Press, 1996.

Requested Resources

- Lecture Room: Loose Seated